# A Multi-Level Visual Analytics Approach to Artist-Era Alignment in Popular Music

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### Why it matters

Popular music evolves over time, but it remains difficult to understand how individual artists are positioned relative to the stylistic norms of their era.

### Research Gap

Prior work has mainly focused on chart prediction [1, 2], aggregate temporal trends [3], or exploration based on embeddings and artist relationships [4].

**Research Question:** How can we visualize how artists **align with or diverge from era-specific stylistic norms** over time?
**Core Idea:** We separate **artist-era alignment** into **shape similarity** and **contrast ratio** to capture stylistic direction and intensity.

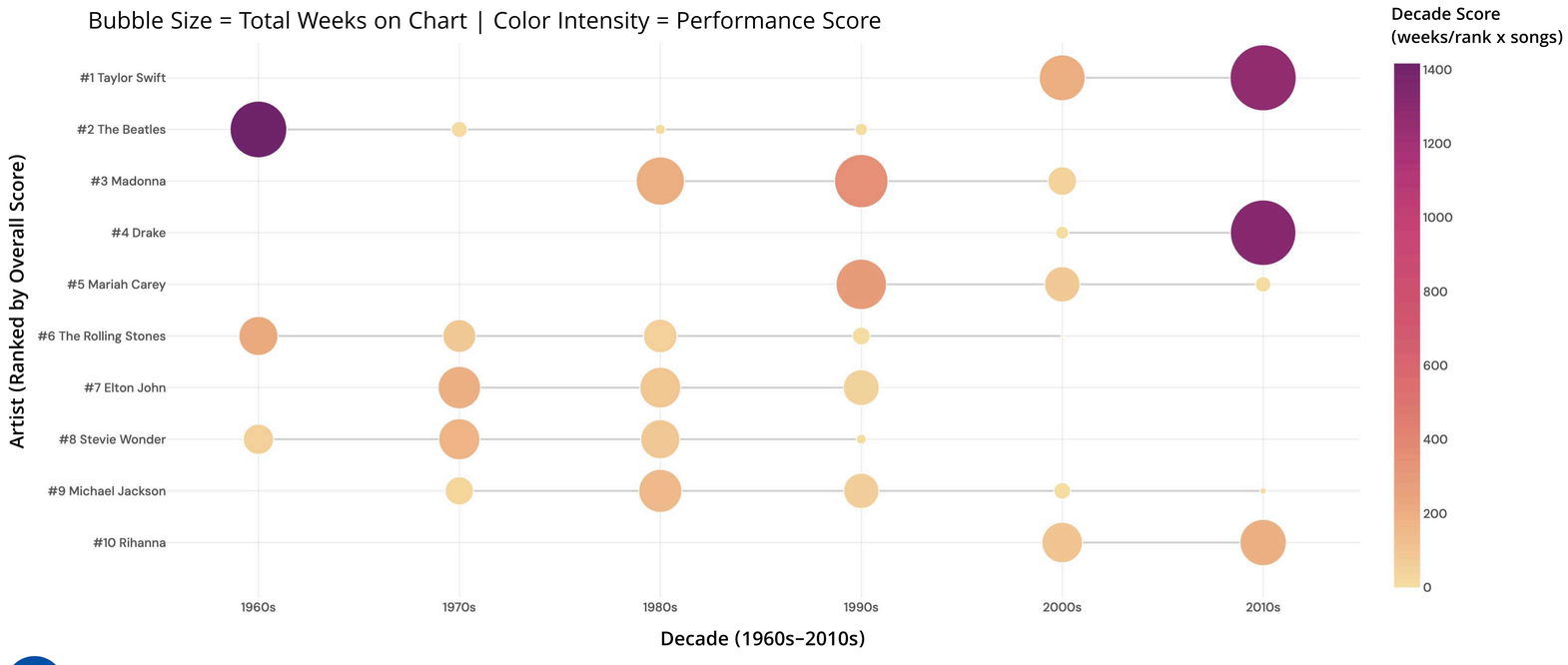
## System Overview

**Dataset:** Weekly U.S. Billboard Hot 100 chart data, Spotify audio features

### 1 Artist–Decade Bubble Chart

**How Did Top Artists' Success and Styles Change Across Six Decades (1960s–2010s)?**

Bubble Size = Total Weeks on Chart | Color Intensity = Performance Score




### 2 Quadrant Trajectory Map

**Artist–Era Quadrant Trajectory Map**

X = Shape Similarity (centered cosine) | Y = Contrast Ratio (σ_artist / σ_era) | Lines = decade-to-decade movement




**Amplified Conformist** Same pattern, more extreme peaks
**Smoothed Conformist** Same pattern, more uniform
**Polarized Maverick** Different pattern, extreme peaks
**Muted Maverick** Different pattern, flat profile

- **Shape similarity** indicates how closely an artist aligns with the era baseline.
- **Contrast ratio** indicates how strongly or smoothly the artist's style is expressed
- The two axes capture **distinct dimensions** (r = –0.19, p = .30).

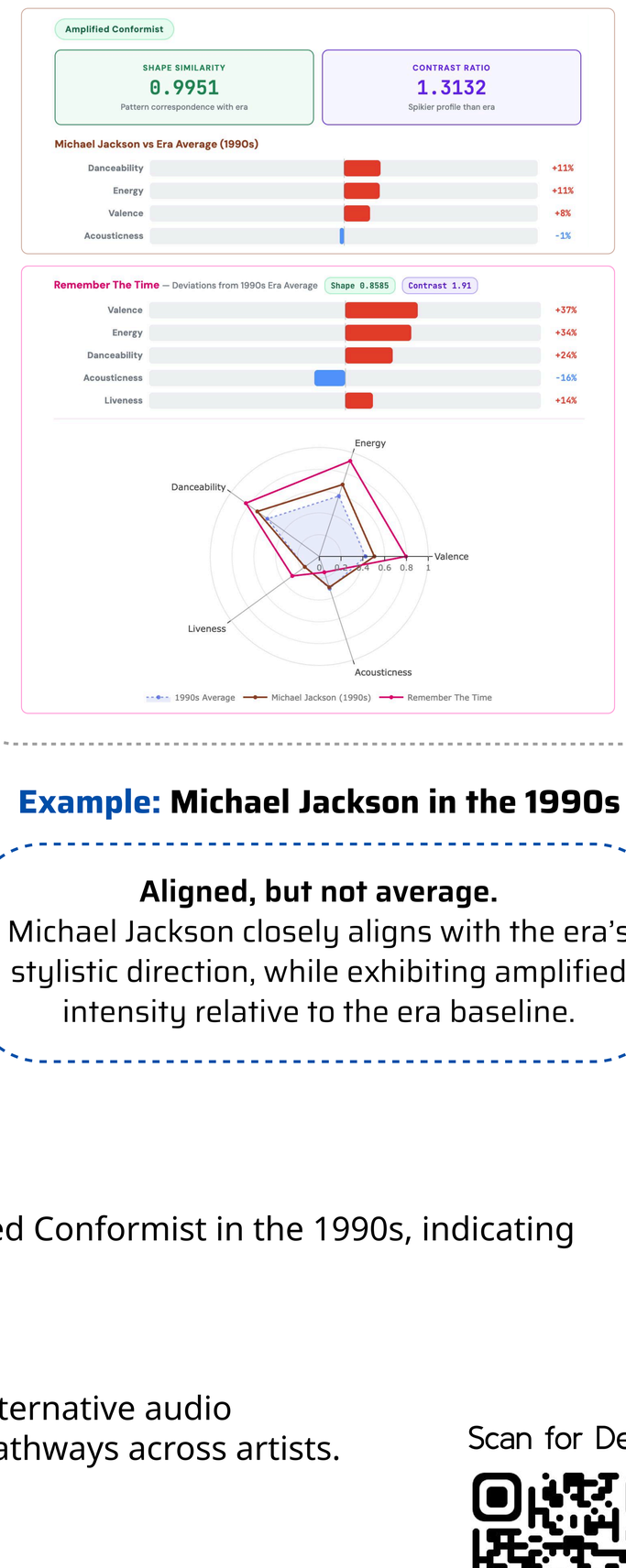
## Details on Demand from Bubble Selection

### 3 Song Performance Table

**Artist:** Michael Jackson
**Decade:** 1990s
**Total Songs:** 11
**Total Weeks on Chart:** 178

| SONG | AVG RANK | PEAK | WEEKS |
|---|---|---|---|
| Remember The Time | 19.00 | 3 | 20 |
| You Are Not Alone | 22.45 | 1 | 20 |
| Black Or White | 24.95 | 1 | 20 |
| Will You Be There | 27.00 | 7 | 20 |
| In The Closet | 40.25 | 6 | 20 |
| Who Is It | 44.28 | 14 | 18 |
| Jam | 46.14 | 26 | 14 |
| Heal The World | 46.70 | 27 | 20 |
| They Don't Care About Us | 54.92 | 30 | 13 |
| Blood On The Dance Floor | 68.73 | 42 | 11 |
| Stranger In Moscow | 94.00 | 91 | 2 |

### 4 Audio Features Profile




**Example: Michael Jackson in the 1990s**

**Aligned, but not average.**
Michael Jackson closely aligns with the era's stylistic direction, while exhibiting amplified intensity relative to the era baseline.

## Takeaways

1. **Alignment and intensity can diverge.**
   An artist can closely follow the stylistic direction of an era while appearing either more amplified or more smoothed relative to the era norm.
2. **Quadrant trajectories reveal measurable career-level transitions.**
   For example, Michael Jackson shifts from near the quadrant boundary in the 1980s to Amplified Conformist in the 1990s, indicating stronger stylistic intensity despite reduced chart activity.

## Future Work

We plan to evaluate the framework with multiple participants and larger datasets, extend it to alternative audio descriptors and sub-decade intervals, and mine trajectory patterns to identify common career pathways across artists.

Scan for Demo

### Reference

**[1]** K. Li, "Predicting Song Popularity in the Digital Age Through Spotify's Data", Theoretical and Natural Science, vol. 39, pp. 68–75, 2024.
**[2]** C. V. Soares Araujo, M. A. Pinheiro de Cristo, and R. Giusti, "Predicting Music Popularity Using Music Charts", IEEE ICMLA 2019.
**[3]** M. Mauch, R. M. MacCallum, M. Levy, and A. M. L. Leroi, "The Evolution of Popular Music: USA 1960–2010", Royal Society Open Science, 2015.
**[4]** F. Levesque and T. Hurtut, "MuzLink: Connected Beeswarm Timelines for Visual Analysis of Musical Adaptations and Artist Relationships", Information Visualization, 2021.